

Baekseok Kim

+1-725-275-1910 | kimb52@unlv.nevada.edu | Portfolio

Skills

Mechanical Design: Mechanisms, actuators and passive joints, mechanical drawings, sensor/electronics integration, geometric-error analysis

CAD & Analysis: Creo, SolidWorks, CATIA, Inventor, Ansys, dynamics, motor torque-speed analysis

Prototyping & Test: Fabrication, assembly, FARO laser tracking, repeatability/payload/reliability testing

Robotics & Software: FK/IK, calibration, EtherCAT integration, motion control, Python, C/C++, MATLAB

Experience

Robotics Engineer / Project Manager

Dec 2018 – Feb 2022

Global Zeus (ZEUS Co., Ltd.) – Republic of Korea

- Led mechanical development of the company's first industrial delta robot, including actuators, passive joints and structural components, from CAD and analysis through fabrication and testing.
- Designed joint modules targeting IP67 and lubrication-free passive joints, plus a SCARA Z-roll module; developed FK/IK and calibration models to compensate for geometric errors.
- Reduced peak torque by more than 10% for the same test motion using motor torque-speed-aware trajectories; evaluated repeatability, payload and reliability with FARO measurement.

Research Assistant

Mar 2022 – Present

University of Nevada, Las Vegas – Las Vegas, NV

- Developed bipedal/humanoid platforms, including the Stilt-bot passive-ankle robot, integrating mechanical hardware with actuators, sensors and control software.
- Led whole-system research in humanoid loco-manipulation and tele-collaboration; served as lead control engineer for Team Avatar-Hubo, an ANA Avatar XPRIZE finalist.

Research Assistant

Mar 2016 – Feb 2019

Hanyang University – Republic of Korea

- Designed a skiing humanoid platform and MATLAB stability simulator, plus mechanisms for segmented rescue robots and interactive exhibition robots.

Selected Publications

- **B. Kim** and P. Oh. "A Hierarchical Control Architecture for Humanoid Loco-Manipulation with Pre-IK Residual RL." *ICSEng*, 2026. **Best Paper Award.**
- **B. Kim** and P. Oh. "Force-augmented LIPM-MPC for Stable Humanoid Loco-Manipulation." *American Control Conference (ACC)*, 2026.
- **B. Kim** and P. Y. Oh. "A Simplified Underactuated Platform for AI-Ready Bipedal Walking Control: The Stilt-bot." *ACC*, pp. 3782–3787, 2025.
- C. Park, **B. Kim**, et al. "Carved Turn Control with Gate Vision Recognition of a Humanoid Robot for Giant Slalom Skiing on Ski Slopes." *Sensors*, 22(3), 816, 2022.

Education

University of Nevada, Las Vegas – PhD, Mechanical Engineering

Expected Dec 2026

Advisor: Paul Y. Oh; GPA: 4.0/4.0

Hanyang University – MS, Interdisciplinary Robot Engineering Systems

Feb 2019

GPA: 4.0/4.0

Sungkyunkwan University – BS, Mechanical Engineering

Feb 2017

Honors & Patent

UKC 2023 Young Scholar Award; co-inventor, delta robot patent application KR20210042030A.